

Pantheon: Canonical Artifact Auditing for Research-Agent Replication

Anonymous research prototype

Abstract

Autonomous research agents can propose hypotheses, modify code, execute experiments, and write scientific claims, but agreement between two runs is not itself evidence of independent replication. Pantheon freezes a canonical research package, executes proposer and replicator runs in isolated workspaces, records artifact provenance, and adjudicates disagreements using canonical hashes, hidden-state canaries, and numerical comparisons. On a controlled benchmark with twenty independently generated packages and eight conditions, Pantheon localizes 160/160 injected faults, compared with 100/160 for pairwise artifact comparison and 20/160 for numerical comparison alone. A cross-implementation public-data case study shows direction agreement on all ten prespecified splits. These experiments validate protocol mechanisms rather than autonomous-agent reliability; no external LLM agents or published research package are replicated here.

1 Introduction

Research-agent systems increasingly automate scientific workflows (Lu et al., 2024; Schmidgall et al., 2025). Benchmarks such as CORE-Bench, PaperBench, and AutoExperiment evaluate whether agents can reproduce or replicate research tasks (Siegel et al., 2024; Starace et al., 2025; Kim et al., 2026). Pantheon asks a complementary question: when two research runs agree or disagree, what artifact evidence supports a replication judgment?

Pairwise agreement can be misleading when both runs share the same bug. Pantheon therefore uses a frozen canonical package as a third reference point and records data, code, protocol, metric, and environment provenance.

2 Method

A claim is represented as $c = (H, P, D, M, E, K)$ for hypothesis, protocol, dataset, metric, expected claim boundary, and code/environment provenance. Each run executes a copied package in a fresh temporary workspace and freezes metrics and hashes. For proposer effect $\hat{\delta}_P$ and replicator effect $\hat{\delta}_R$, standardized disagreement is

$$D_z = \frac{|\hat{\delta}_P - \hat{\delta}_R|}{\sqrt{s_P^2 + s_R^2 + \epsilon}}.$$

The rule adjudicator checks canary exposure and canonical artifact drift before numerical disagreement.

3 Experiments

The controlled benchmark uses twenty synthetic datasets generated from $y = 0.8x + 1.25t + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$. Eight conditions are evaluated: clean, data mismatch, metric mismatch, implementation mismatch, environment mismatch, protocol ambiguity, hidden-state contamination, and a correlated shared implementation bug. Three methods compare the same frozen run pairs: numeric-only, pairwise artifacts, and full Pantheon.

A public-data case study uses the Breast Cancer Wisconsin Diagnostic dataset bundled with scikit-learn. Across ten prespecified stratified splits, a scikit-learn L-BFGS logistic-regression proposer is compared with an independently implemented numpy batch-gradient-descent replicator.

4 Results

Method	Correct	Accuracy
Numeric only	20/160	0.125
Pairwise artifacts	100/160	0.625
Full Pantheon	160/160	1.000

Table 1: Controlled fault localization. The benchmark is engineered and should not be interpreted as real-world accuracy.

In the shared-bug condition, both weaker baselines label all twenty comparisons as reproduced because proposer and replicator use identical corrupted code. Pantheon detects that both differ from the frozen canonical implementation. In the public case, both implementations improve over majority accuracy on 10/10 splits; their mean absolute difference in the improvement metric is 0.00699 accuracy points.

5 Limitations

No external LLM agents are evaluated. No published paper package is independently replicated. The fault taxonomy is engineered around observable provenance channels, byte-level hashes are not semantic equivalence tests, and workspace/process isolation is weaker than container/host/model-provider isolation.

6 Conclusion

Canonical artifact auditing provides information that pairwise agreement alone cannot. The next decisive experiment is to apply Pantheon to genuine research-agent trajectories and third-party research packages with blinded replication and stronger isolation.

References

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